

RAED OUIRIEMMI

Software Developer / M.Sc. in Data Science

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PROFILE

I hold a Master's in Data Science and a Bachelor's in Software Engineering. Over the past two years I have built several applications that combine a web or mobile frontend with a Python backend and some AI component (language models, NLP, or computer vision). I am comfortable working alone on a project from the first commit to the server it runs on. I am currently looking for a developer or data role based in Canada.

TECHNICAL SKILLS

Programming:	Python, JavaScript, Java, C#, C++
Web & Mobile:	React, React Native (Expo), Node.js, FastAPI, HTML, CSS, Tailwind
Machine Learning:	TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Deep Learning & AI:	PyTorch, Keras, YOLO, LLaMA 2, BART, Helsinki-NLP, OpenAI API, OpenRouter API
Databases:	PostgreSQL, MySQL, MSSQL, MongoDB, Firebase, Supabase (with SQLAlchemy and Alembic)
DevOps:	Docker, Docker Compose, nginx, Linux / Ubuntu, UFW, fail2ban, Git, GitHub
Other tools:	Maven, JUnit, Jupyter, Android Studio, Visual Studio
Methods:	UML, Agile, testing, web scraping, localization (English / French / Arabic RTL)

WORK EXPERIENCE

AI / Full-Stack Engineer Intern, AKR Smart Consulting

Feb 2026 - May 2026

Master's final-year project (PFE)

- Built a multi-tenant AI chatbot platform (SaaS). Admins create assistants trained on their own content, such as uploaded documents and crawled websites, using a RAG pipeline with sentence-transformers embeddings and ChromaDB vector search.
- Delivered a FastAPI + MongoDB backend, a React admin dashboard, an embeddable JavaScript chat widget, and a Celery / Redis / Playwright web crawler.
- JWT authentication with 2FA (TOTP), per-IP rate limiting, and Dockerized deployment.

Software Development Intern, AZ IT

Feb 2024 - Aug 2024

Monastir, Tunisia

- Six-month internship where I worked on two projects back to back: a mobile app for accountants and a web platform for business addresses.

Mobile app for accountants and clients (React Native)

Jun 2024 - Aug 2024

- Built a mobile app and a small web dashboard so accountants and their clients could share documents without using email.
- Added secure file upload and a dashboard where the accountant can see all clients at a glance.

Business address platform (graduation project)

Feb 2024 - May 2024

- Final-year project with AZ IT. A web app that lets small companies register a legal office address without renting a physical office.
- Did the frontend in React.js and the backend in Node.js, plus the address declaration form.

PROJECTS

AppFullSport, a fitness coaching app

2025 - 2026

- Personal project. A mobile app for people who work out, with a React Native client, a FastAPI backend and a small React admin panel. The whole thing runs in Docker on a VPS I manage myself.
- Workout sessions with weight and rep tracking per set, a rest timer, and automatic detection of personal records. I seeded the database with around 83 exercises.
- Step counting works on both platforms: on iOS through HealthKit and CoreMotion, on Android through the hardware step sensor.
- Spent a good amount of time on the server side: UFW and DOCKER-USER iptables rules, fail2ban, SSH keys only, rate limits at both nginx and FastAPI level, CSP and HSTS headers, rotated secrets. Schema changes are handled with Alembic migrations.
- The app is fully translated into English, French and Arabic (right to left). Even backend error messages are translated based on the Accept-Language header.
- Stack: React Native (Expo SDK 54), FastAPI, SQLAlchemy, Alembic, PostgreSQL 16, Docker Compose, nginx, React, Tailwind.

Chatbot that answers questions from an SQL database

2025

- Personal project. A small web chatbot where you ask a question in plain English and it runs the right SQL query against an MSSQL database and gives you back a short answer.
- Stack: React, FastAPI, MSSQL, OpenRouter API.

Discord multi-account management tool

2025

- Web app that manages several Discord accounts with tokens and proxies. There is an admin dashboard to follow what each account is doing in real time.

- The bots can reply on their own using the OpenAI API, and messages are sent through Discord webhooks.
- Stack: React, FastAPI, Discord webhooks, OpenAI API.

IT product search chatbot with LLaMA 2

2024

- Chatbot that helps the user find laptops, phones and other IT products. It uses LLaMA 2 to understand the question and a scraper that pulls the catalog from MyTek once a day.
- Gives personal suggestions based on budget and use case.

Text summarization and translation web app

2024

- Web app built with FastAPI and React. Uses BART for summarization and Helsinki-NLP for translation. Supports more than 10 languages.

License plate detection in videos (YOLO)

Sep 2024

- Trained a YOLO model to find and extract car license plates from a video, frame by frame, in real time.

EDUCATION

Master's in Data Science

Sep 2024 - Jul 2026

Faculty of Sciences of Monastir, Tunisia

Bachelor's in Software Engineering and Information Systems

2021 - 2024

Faculty of Sciences of Monastir, Tunisia

Scientific Baccalaureate

2021

Dhouiher High School, Zarzis, Tunisia

LANGUAGES

Arabic: native

French: upper intermediate (TCF 400/699, B2)

English: intermediate (B1, I can work in English on a daily basis)

PERSONAL QUALITIES

Problem solving, critical thinking, adaptability, leadership, creativity, teamwork.